

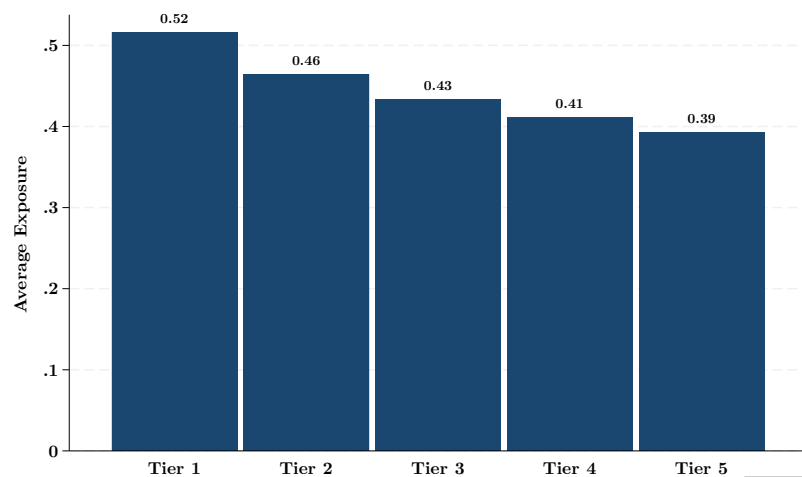


Figure A.15: Average Exposure by School Quality (Juniors, 2022)

Notes: Bars report exposure for juniors employed in 2022, by university prestige category. The standard deviation of the exposure variables is 0.21.

A.4 Robustness of Adoption Measure: Postings and Worker Position Descriptions

The baseline adoption indicator Adopt_i in the main text is identified from explicit “GenAI integrator” job postings (Section 4.2). This appendix re-estimates the headline triple-difference of Section 5.2.2 using an *augmented* adoption indicator, $\text{Adopt}_i^{\text{aug}}$, that flags firm i as an adopter if it either (a) appears as an adopter under the baseline postings-based definition, or (b) has at least one worker whose own LinkedIn *position* description is classified as an integrator role. The augmented measure helps address two residual concerns about the baseline definition. First, a fraction of GenAI integrator postings may be “ghost” or signaling postings that never translate into on-the-job integration work. Second, “silent adopters”—firms that integrate GenAI through internal reskilling or in-house tooling rather than through external hiring—can be missed by a postings-only definition.

The position-description component is constructed in close parallel to the postings-based measure (see Section 4.2 for the full pipeline): we first filter the 2021–2025 worker position spells to those whose free-text description contains at least one GenAI-related keyword, and we then ask an LLM (Llama-3.3-70B-Instruct, run with temperature = 0 and a fixed seed) to keep only those whose description indicates that the worker integrates GenAI into a product, workflow, or pipeline. A firm is flagged by component (b) above if it has at least one such position.

Across the 281,898 firms in the analysis sample, the postings-based definition flags 10,433 adopters (3.71%). Adding the position-description component flags an additional 34,387 firms not reached by the postings data, for a union of 38,441 augmented adopters (13.64%). The augmented measure thus significantly expands the treated group relative to the baseline. Figure A.16 reports the triple-difference event-study profile from Equation 11, replacing Adopt_i with $\text{Adopt}_i^{\text{aug}}$. The pattern is qualitatively unchanged: The magnitude is very similar to the baseline estimate of about -0.09 in Figure 5. Even when the treated group is expanded to include firms whose adoption is visible only through workers’ own descriptions of what they do, the seniority-biased contraction in junior employment after the launch of ChatGPT remains intact.

It is worth noting that the position-description data are considerably less complete and less informative than the job-posting texts on which the baseline measure is built. Only

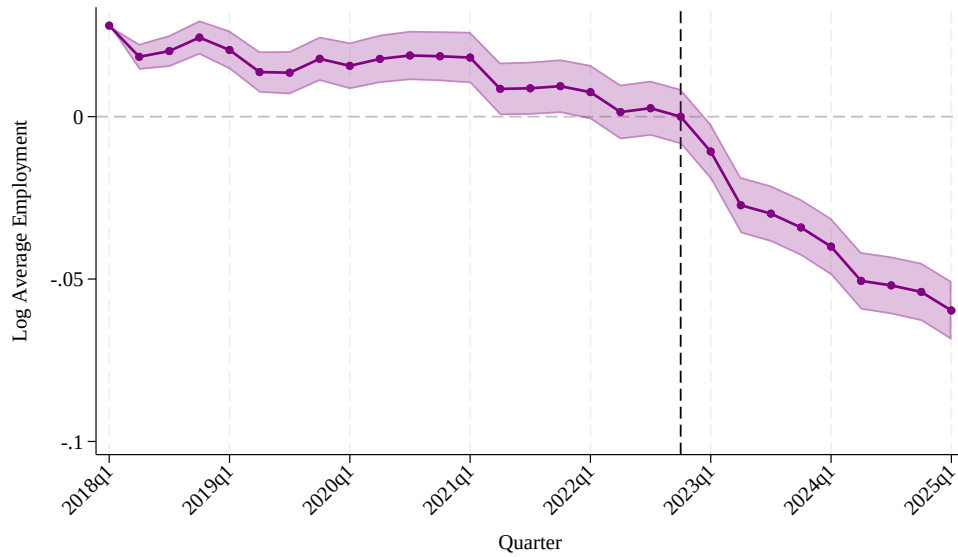


Figure A.16: Triple-Difference Estimates—Augmented Adoption Measure

Notes: This figure reproduces Figure 5, replacing the postings-based adoption indicator Adopt_i with the augmented indicator $\text{Adopt}_i^{\text{aug}}$, which additionally treats a firm as an adopter if any of its worker position descriptions in the 2021–2025 window is classified as an integrator role by Llama-3.3-70B-Instruct. Specification, fixed effects, and inference follow Equation 11. Standard errors are clustered at the firm level.

about 42% of U.S. position spells in the 2021–2025 window carry a free-text description at all, and those that do are often short or formulaic.

A.5 Theoretical Framework: Automation Tomorrow, Hiring Reductions Today

This section develops a parsimonious framework in which expectations of *future* labor-saving productivity gains depress *current* hiring when employment is costly to adjust. We first present a two-period model that delivers an explicit “firing wedge” in the period-0 hiring condition. We then embed the same mechanism in a task-based production structure in the spirit of [Acemoglu and Restrepo \(2019\)](#).

A.5.1 A Baseline Two-Period Model

Time is $t \in \{0, 1\}$. A representative firm chooses employment n_t each period and produces with a concave, increasing production function $f(\cdot)$; per-worker wages w_t are exogenous. The firm discounts with factor $\beta \in (0, 1]$ and faces a per-unit hiring cost $c_h \geq 0$ in $t = 0$. Reductions in employment incur a per-worker firing cost $\phi > 0$ on separated workers, so that $\Phi = \phi(n_0 - n_1)_+$. Let $A_t > 0$ denote period- t productivity. Anticipated labor-saving progress in period 1 is captured by a higher A_1 that lowers the amount of labor required per unit of output.

The firm’s objective is

$$\max_{n_0, n_1} \Pi = A_0 f(n_0) - (w_0 + c_h)n_0 + \beta \mathbb{E}_0[A_1 f(n_1) - w_1 n_1 - \phi(n_0 - n_1)_+]. \quad (16)$$

In period 1, conditional on n_0 , the first-order condition (FOC) equates the marginal product of labor (MPL) to the effective marginal cost:

$$A_1 f'(n_1) = \begin{cases} w_1, & \text{if } n_1 \geq n_0 \text{ (no firing),} \\ w_1 - \phi, & \text{if } n_1 < n_0 \text{ (firing).} \end{cases}$$

In period 0, the hiring decision internalizes that raising n_0 today can trigger firing tomorrow. The FOC is

$$A_0 f'(n_0) = \begin{cases} w_0 + c_h, & \text{if no firing is expected,} \\ w_0 + c_h + \beta\phi, & \text{if firing is anticipated,} \end{cases} \quad (17)$$